Jack Schulte | Curriculum Vitae

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Education

2021 – present ◇ **Ph.D. Astrophysics & Astronomy**, Michigan State University
Advisor: Dr. Joseph Rodriguez
Area of Interest: Formation and evolution of hot Jupiters

2017 – 2021 ◇ **B.S. Physics**, Arizona State University
Advisor: Dr. Maitrayee Bose
Area of Interest: Origins of presolar grains
Minor: Astrophysics
Summa Cum Laude
Barrett, The Honors College

Research Interests

- **Giant planet formation and evolution**: Discovering and characterizing hot Jupiters in order to construct a self-consistent sample of system parameters to constrain the origin channels of hot Jupiters and the long-term stability of planetary systems
- **Planet engulfment**: Finding signatures of engulfed planets in subgiant stars and connecting engulfment events to giant planet migration
- **Presolar grains as tracers**: Using presolar grains and models of stellar evolution and nucleosynthesis to glean information about the solar neighborhood 4.6 billion years ago

UAT Keywords: Hot Jupiters (753), Exoplanet migration (2205), Exoplanet catalogs (488)

Teaching

Spring 2023 ◇ **AST 208: Planets and Telescopes**, Michigan State University
I acted as a grader for this astronomy course designed for astronomy undergraduates. I developed a rubric system to expedite the grading of submitted Jupyter notebooks. In addition, I hosted weekly office hours and led several lab sessions.

Spring 2022 – Fall 2022 ◇ **ISP 205L: Visions of the Universe**, Michigan State University
Each semester, I was the primary instructor for one section and the secondary instructor for another section of this astronomy lab course designed for non-STEM majors. In each class, I presented a 20-30 minute mini-lecture. I also redesigned several labs and designed and ran planetarium shows. Class sizes ranged from 50-100 students.

Fall 2021 ◇ **PHY 191: Physics Laboratory for Scientists**, Michigan State University
I led three sections, each containing 20 students, in a weekly three-hour lab designed to teach the fundamentals of physics (mechanics) to science and engineering undergraduates. This involved preparing 10-20 minute chalkboard mini-lectures followed by a hands-on demonstration of the lab.

Fellowships

May 2026 – December 2026 ◇ **NASA Michigan Space Grant Consortium Graduate Fellowship**
Title: "The Migration and Evolution of Giant Exoplanets"
Amount: $5,000

May 2026 – August 2026 ◇ **College of Natural Science Dissertation Completion Fellowship**
Amount: $9,600

May 2025 – December 2025 ◇ **NASA Michigan Space Grant Consortium Graduate Fellowship**
Title: "The Migration and Evolution of Giant Exoplanets"
Amount: $5,000

Positions

2021 – present ◇ **Graduate Teaching/Research Assistant**, Michigan State University
Advisor: Dr. Joseph Rodriguez
Leading the Migration and Evolution of giant ExoPlanets (MEEP) survey to discover and characterize all remaining hot Jupiters orbiting bright Sun-like stars with NASA's TESS mission. I also have spent over 300 hours performing spectroscopy and photometry with the Goodman High-Throughput Spectrograph and the SPARTAN infrared imager on the SOAR 4.1-meter telescope.

2019 – 2021 ◇ **Research Aide**, Lunar Reconnaissance Orbiter Camera
Advisor: Dr. Mark Robinson
Built spatiotemporal maps to track the movement and activities of the astronauts in each of the Apollo missions. Built `python` scripts to automate much of the work.

2018 – 2021 ◇ **Undergraduate Research Assistant**, Arizona State University
Advisor: Dr. Maitrayee Bose
Used 3D core-collapse supernova models to constrain the origins of presolar stardust grains.

Students Advised

2025-present ◇ **Benjamin Navock**, Undergraduate Student, Michigan State University
Leading the discovery of several giant planet systems with wide binary companions.

2024-2025 ◇ **Isaac Fournier**, Undergraduate Student, Michigan State University
Built a custom pipeline to search through *TESS* data for hot Jupiters orbiting K dwarf stars.

Publications

Total refereed citations: 170+ | h-index: 7

Peer-Reviewed Journal Articles

1. B. Kotten et al., "Lithium Enrichment in a Subgiant Star with a Brown Dwarf Companion: A Planetary Engulfment Candidate," *The Astrophysical Journal*, vol. 1004, no. 2, p. 193, Jun. 2026, Art. no. 193. DOI: 10.3847/1538-4357/ae71bb arXiv: 2605.21407 [astro-ph.SR].

2. **J. Schulte** et al., "Migration and Evolution of giant ExoPlanets (MEEP). III. Twenty-Nine Giant Planets from the TESS Mission," *arXiv:2609.38468*, Sep. 2026. DOI: 10.48550/arXiv.2609.38468 Accessed: Oct. 1, 2026.

3. J. D. Simon et al., "TOI-7169 b: A Hot Jupiter Transiting a Metal-poor Star," *The Astronomical Journal*, vol. 172, no. 2, p. 118, Aug. 2026, Art. no. 118. DOI: 10.3847/1538-3881/ae7f02 arXiv: 2603.25787 [astro-ph.EP].

4. A. Bieryla et al., "TOI-2005b: An Eccentric Warm Jupiter in Spin-orbit Alignment," *The Astronomical Journal*, vol. 169, no. 5, p. 273, May 2025, Art. no. 273. DOI: 10.3847/1538-3881/adc441 arXiv: 2503.20069 [astro-ph.EP].

5. A. Distler et al., "TESS Hunt for Young and Maturing Exoplanets (THYME). XII. A Young Mini-Neptune on the Upper Edge of the Radius Valley in the Hyades Cluster," *The Astronomical Journal*, vol. 169, no. 3, p. 166, Mar. 2025, Art. no. 166. DOI: 10.3847/1538-3881/ada26a arXiv: 2410.11990 [astro-ph.EP].

6. **J. Schulte** et al., "Migration and Evolution of giant ExoPlanets (MEEP). II. Super-Jupiters and Lithium-rich host stars," *Monthly Notices of the Royal Astronomical Society*, vol. 543, no. 1, pp. 292–313, Oct. 2025. DOI: 10.1093/mnras/staf1455 arXiv: 2509.02666 [astro-ph.EP].

7. N. Vowell et al., "Eleven New Transiting Brown Dwarfs and Very-low-mass Stars from TESS," *The Astronomical Journal*, vol. 170, no. 2, p. 68, Aug. 2025, Art. no. 68. DOI: 10.3847/1538-3881/addd17 arXiv: 2501.09795 [astro-ph.EP].

8. S. W. Yee et al., "The TESS Grand Unified Hot Jupiter Survey. III. Thirty More Giant Planets," *The Astrophysical Journal Supplement*, vol. 280, no. 1, p. 30, Sep. 2025, Art. no. 30. DOI: 10.3847/1538-4365/aded0d arXiv: 2507.01855 [astro-ph.EP].

9. **J. Schulte** et al., "Migration and Evolution of giant ExoPlanets (MEEP). I. Nine Newly Confirmed Hot Jupiters from the TESS Mission," *The Astronomical Journal*, vol. 168, no. 1, p. 32, Jul. 2024, Art. no. 32. DOI: 10.3847/1538-3881/ad4a57 arXiv: 2401.05923 [astro-ph.EP].

10. E. A. Gilbert et al., "A Second Earth-sized Planet in the Habitable Zone of the M Dwarf, TOI-700," *The Astrophysical Journal Letters*, vol. 944, no. 2, p. L35, Feb. 2023, Art. no. L35. DOI: 10.3847/2041-8213/acb599 arXiv: 2301.03617 [astro-ph.EP].

11. J. E. Rodriguez et al., "Another shipment of six short-period giant planets from TESS," *Monthly Notices of the Royal Astronomical Society*, vol. 521, no. 2, pp. 2765–2785, May 2023. DOI: 10.1093/mnras/stad595 arXiv: 2205.05709 [astro-ph.EP].

12. S. P. Schmidt et al., "Verification of Gaia Data Release 3 Single-lined Spectroscopic Binary Solutions With Three Transiting Low-mass Secondaries," *The Astronomical Journal*, vol. 166, no. 6, p. 225, Dec. 2023, Art. no. 225. DOI: 10.3847/1538-3881/ad0135 arXiv: 2310.07936 [astro-ph.SR].

13. T. W. Carmichael et al., "TOI-2119: a transiting brown dwarf orbiting an active M-dwarf from NASA's TESS mission," *Monthly Notices of the Royal Astronomical Society*, vol. 514, no. 4, pp. 4944–4957, Aug. 2022. DOI: 10.1093/mnras/stac1666 arXiv: 2202.08842 [astro-ph.SR].

14. **J. Schulte**, M. Bose, P. A. Young, and G. S. Vance, "Three-dimensional Supernova Models Provide New Insights into the Origins of Stardust," *The Astrophysical Journal*, vol. 908, no. 1, p. 38, Feb. 2021, Art. no. 38. DOI: 10.3847/1538-4357/abcd41 arXiv: 2011.07459 [astro-ph.HE].

Conference Proceedings

1. **J. Schulte** and J. Rodriguez, "The Migration and Evolution of giant ExoPlanets (MEEP) Survey," in *American Astronomical Society Meeting Abstracts*, ser. American Astronomical Society Meeting Abstracts, vol. 247, Feb. 2026, 349.02, p. 349.02.

2. **J. Schulte**, "Understanding the Migration History of Hot Jupiters," in *TESS Science Conference III*, Jul. 2024, 52, p. 52. DOI: 10.5281/zenodo.13118111

3. **J. Schulte** and J. Rodriguez, "The Migration and Evolution of Eccentric Planets (MEEP) survey," in *AASTCS10, Extreme Solar Systems V*, ser. AAS/Division for Extreme Solar Systems Abstracts, vol. 56, Apr. 2024, 611.07, p. 611.07.

4. **J. Schulte** and J. Rodriguez, "The Migration and Evolution of Eccentric Planets (MEEP) Survey," in *AAS/Division of Dynamical Astronomy Meeting*, ser. AAS/Division of Dynamical Astronomy Meeting, vol. 55, Sep. 2023, 205.09, p. 205.09.

5. N. R. Gonzales et al., "Pedal to the Metal: Apollo 15 Spatiotemporal Mapping of Act II of Manned Lunar Exploration," in *53rd Lunar and Planetary Science Conference*, ser. LPI Contributions, vol. 2678, Mar. 2022, 2672, p. 2672.

6. M. Bose, **J. Schulte**, G. Vance, R. A. Jansen, and P. Young, "Heterogeneous R-Process Chromium and Titanium Ejecta from Core Collapse Supernova Ejecta Polluted Our Solar System," in *52nd Lunar and Planetary Science Conference*, ser. Lunar and Planetary Science Conference, Mar. 2021, 1414, p. 1414.

7. N. R. Gonzales, **J. A. Schulte**, and M. S. Robinson, "In the Footsteps of the First: Apollo 14 Spatiotemporal Map," in *5th Planetary Data Workshop & Planetary Science Informatics & Analytics*, ser. LPI Contributions, vol. 2549, Jun. 2021, 7062, p. 7062.

8. M. Bose, S. Starrfield, P. A. Young, G. Vance, and **J. Schulte**, "Origins of O-Anomalous Stardust Using New Nova and Supernova Modeling," in *51st Annual Lunar and Planetary Science Conference*, ser. Lunar and Planetary Science Conference, Mar. 2020, 1117, p. 1117.

9. N. R. Gonzales, **J. A. Schulte**, M. R. Henriksen, R. V. Wagner, and M. S. Robinson, "Tremors and Tracks: Tracing the Apollo 12 Astronauts Through Time," in *51st Annual Lunar and Planetary Science Conference*, ser. Lunar and Planetary Science Conference, Mar. 2020, 1578, p. 1578.

10. **J. Schulte**, M. Bose, P. Young, and G. Vance, "The Supernova Origins of Rare Stardust Enriched with 13C and 15N," in *51st Annual Lunar and Planetary Science Conference*, ser. Lunar and Planetary Science Conference, Mar. 2020, 1268, p. 1268.

11. **J. Schulte**, M. Bose, P. Young, and G. Vance, "Using Symmetric and Asymmetric Three-Dimensional Supernova Models to Constrain the Origins of Presolar SiC Grains," in *50th Annual Lunar and Planetary Science Conference*, ser. Lunar and Planetary Science Conference, Mar. 2019, 1746, p. 1746.

Invited Talks

October 2025 ◇ **SPLAT Talk Series**, University of Hawaii

March 2025 ◇ **Michigan State University Astronomy Seminar**, Michigan State University

May 2023 ◇ **JPL Virtual Exoplanet Lecture Series**, NASA Jet Propulsion Laboratory

Conference Presentations

January 2026 ◇ **247th Meeting of the American Astronomical Society**: Phoenix, AZ (contributed talk)

November 2025 ◇ **Great Lakes Exoplanet Area Meeting**: Madison, WI (contributed talk)

October 2025 ◇ **Michigan Space Grant 2025 Fall Conference**: Mt. Pleasant, MI (contributed talk)

June 2025 ◇ **Emerging Researchers in Exoplanet Science X**: Princeton, NJ (contributed talk)

May 2025 ◇ **Third International Symposium of Science with the SOAR Telescope**: East Lansing, MI (contributed talk)

February 2025 ◇ **Know Thy Star 2**: Pasadena, CA (contributed poster)

July 2024 ◇ **TESS Science Conference III**: Cambridge, MA (contributed poster)

◇ **Emerging Researchers in Exoplanet Science IX**: Ithaca, NY (contributed poster)

March 2024 ◇ **AAS Extreme Solar Systems V**: Christchurch, New Zealand (contributed poster)

September 2023 ◇ **Giant Magellan Telescope Community Science Meeting**: Washington D.C. (contributed poster)

June 2023 ◇ **Emerging Researchers in Exoplanet Science VIII**: New Haven, CT (contributed talk)

May 2023 ◇ **AAS Division on Dynamical Astronomy**: East Lansing, MI (contributed poster)

Conference Presentations (continued)

October 2022 ◇ **Great Lakes Exoplanet Area Meeting**: Columbus, OH (contributed talk)

June 2022 ◇ **Emerging Researchers in Exoplanet Science VII**: State College, PA (contributed poster)

April 2020 ◇ **Arizona Space Grant Consortium Statewide Symposium**: Tempe, AZ (contributed talk)

February 2020 ◇ **ASU School of Earth and Space Exploration Research Symposium**: Tempe, AZ (contributed poster)

April 2019 ◇ **Arizona Space Grant Consortium Statewide Symposium**: Tempe, AZ (contributed talk)

March 2019 ◇ **Lunar and Planetary Science Conference**: The Woodlands, TX (contributed talk)

Successful Proposals

2026 ◇ *Investigating the Origins of Hot Jupiters in Wide Binaries* | NOIRLab 2026B Proposal (NEID) | PI: **Jack Schulte**

2025 ◇ *The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution* | JWST Cycle 4 GO Proposal | PI: Peter Gao, CoI: **Jack Schulte**

◇ *Completing the Magnitude-Limited Sample of Hot Jupiters via TESS* | NOIRLab 2025B Proposal (CHIRON) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

◇ *Identifying the Origin of Giant Planet Migration with TESS* | NOIRLab 2025A Proposal (CHIRON) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

◇ *Probing Stellar Obliquity Across the Giant Planet - Brown Dwarf Transition* | NOIRLab 2025A Proposal (NEID) | PI: Noah Vowell, CoI: **Jack Schulte**

2024 ◇ *Identifying the Origin of Giant Planet Migration with TESS* | NOIRLab 2024B Proposal (CHIRON/MINERVA) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

2023 ◇ *Identifying the Origin of Giant Planet Migration with TESS* | NOIRLab 2023B Proposal (CHIRON/MINERVA) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

◇ *Confirmation and Characterization of Dynamically Young Jupiters to Understand Giant Planet Migration* | NOIRLab 2023A Proposal (CHIRON/MINERVA) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

2022 ◇ *Extending the Science Yield of TESS to Cool Jupiters* | NOIRLab 2022A Proposal (CHIRON) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

◇ *Confirmation and Characterization of Dynamically Young Jupiters to Understand Giant Planet Migration* | NOIRLab 2022A Proposal (CHIRON/MINERVA) | PI: Joseph Rodriguez, CoI: **Jack Schulte**

Outreach and Service

- Currently the primary graduate student leader/mentor for the MSU Observatory Research Program, a program designed to give undergraduates experience observing with a 0.6-m telescope. In this program, students perform observations and research that is then contributed to peer-reviewed publications.
- Active leader of MSU Public Observing Nights, which regularly see hundreds of attendees visiting MSU's campus observatory. I give tours of the observatory's dome and telescope and train volunteers on operating the 0.6-m telescope as well as smaller electronic telescopes.
- Mentored two graduate students and one undergraduate student through MSU's Stellar Mentorship program, which connects graduate and undergraduate astronomy students with mentors and mentees at similar stages of their academic careers.

- Member of MSU's Astro Coffee Committee for 4 years, where I sought speakers to present papers to the department twice per week.
- Presented four Astronomy on Tap talks to a general public audience.

Awards

2024 ◇ **Outreach Award** (for public outreach at MSU observatory)

2023 ◇ **Physics and Astronomy Teaching Award**

2022 ◇ **Harlo Mork Graduate TA Excellence in Teaching Award**

2018-2021 ◇ **ASU/NASA Space Grant** ($11,200 over seven semesters to pursue research at the School of Earth and Space Exploration)

2020 ◇ **ASU President's Scholarship** ($15,000 to one student for travel to Antarctica in December 2020)

2019 ◇ **Nininger Student Travel Award** ($1000 for travel to the 50th Lunar and Planetary Science Conference)

References

- Dr. Joseph E. Rodriguez (jrod@msu.edu)
- Dr. Melinda Soares-Furtado (mmsoares@wisc.edu)
- Dr. David W. Latham (dlatham@cfa.harvard.edu)